

# ECN 594: Oligopoly Refresher

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# Plan

1. **Cournot competition (refresher)**
2. Bertrand competition (refresher)
3. Escaping the Bertrand paradox

## From ECN 532: Oligopoly models

- You covered Cournot and Bertrand in Hector's class
- Today: quick refresher in IO notation
- **New focus:** Market power measurement
  - Connecting oligopoly models to Lerner index
  - When does each model apply?

## Cournot competition: setup

- $n$  firms producing homogeneous goods
- Firms choose **quantities** simultaneously
- Inverse demand:  $P = P(Q)$  where  $Q = \sum_{i=1}^n q_i$
- Constant marginal cost:  $c$
- Firm  $i$  profit:  $\pi_i = P(Q) \cdot q_i - c \cdot q_i$

## Worked example: Cournot duopoly

- **Setup:** Inverse demand is  $P = 100 - Q$ . Two firms with  $MC = 10$ .
- **Steps:**
  1. Write down firm  $i$ 's profit function
  2. Find firm  $i$ 's best response function
  3. Use symmetry to find equilibrium  $Q$  and  $P$

## Worked example: Cournot duopoly (solution)

- **Step 1:** Profit:  $\pi_i = (100 - q_i - q_j)q_i - 10q_i$
- **Step 2:** FOC:  $\frac{d\pi_i}{dq_i} = 100 - 2q_i - q_j - 10 = 0$
- Best response:  $q_i = \frac{90 - q_j}{2} = 45 - 0.5q_j$
- **Step 3:** Symmetry:  $q_1 = q_2 = q^*$
- $q^* = 45 - 0.5q^* \Rightarrow 1.5q^* = 45 \Rightarrow q^* = 30$
- $Q = 60, P = 100 - 60 = 40$
- Profit per firm:  $\pi = (40 - 10) \times 30 = 900$

## Cournot with $n$ symmetric firms

- **Setup:**
- $n$  firms with constant marginal cost  $c$
- Linear demand:  $P = a - bQ$  where  $Q = q_1 + q_2 + \dots + q_n$
- **Question:** What are the Cournot equilibrium quantities and price?

## Cournot with $n$ symmetric firms (solution)

- Firm  $i$  profit:  $\pi_i = q_i(a - bQ) - cq_i$
- FOC:  $\frac{d\pi_i}{dq_i} = a - bQ - bq_i - c = 0$
- By symmetry:  $q_1 = q_2 = \dots = q_n = q^*$ , so  $Q = nq^*$
- Substituting:  $a - bnq^* - bq^* - c = 0$
- Solving:  $q^* = \frac{a-c}{b(n+1)}$ ,  $P^* = \frac{a+nc}{n+1}$
- Profit:  $\pi^* = (P^* - c)q^* = \frac{a-c}{n+1} \cdot \frac{a-c}{b(n+1)} = \frac{(a-c)^2}{b(n+1)^2}$
- **Key insight:** As  $n \rightarrow \infty$ ,  $P \rightarrow c$  and  $\pi \rightarrow 0$



## Practice: Cournot comparative statics

- **True, False, or NEI:**
- (a) Adding a third firm to a Cournot duopoly always reduces industry profits.
- (b) In Cournot, all firms must have the same marginal cost.

## Practice: Cournot comparative statics (solution)

- **Answers:**
- **(a) TRUE.** More firms  $\Rightarrow$  lower price  $\Rightarrow$  lower industry profit. Each firm's profit falls, and total profit falls because  $P$  is closer to  $MC$ .
- **(b) FALSE.** Asymmetric costs work fine. Low-cost firms produce more and earn higher margins.

# Plan

1. Cournot competition (refresher)
2. **Bertrand competition (refresher)**
3. Escaping the Bertrand paradox

## Bertrand competition: setup

- $n$  firms producing **homogeneous** goods
- Firms choose **prices** simultaneously
- Consumers buy from lowest-price firm
- If tie: split demand equally
- Constant marginal cost:  $c$

## Bertrand: the paradox

- **Nash equilibrium:**  $p_1 = p_2 = c$  (marginal cost pricing!)
- **Why?**
  - If  $p_i > p_j > c$ : firm  $i$  can undercut and capture entire market
  - Undercutting continues until  $p = c$
- **The “paradox”:**
  - Only 2 firms, but competitive outcome!
  - Zero profits with just 2 competitors
  - Seems unrealistic for most markets

## Potential solutions to the Bertrand paradox

- How could we change the assumptions of the Bertrand model to get a more realistic model with positive profits (i.e. firms pricing above MC)?
- **Asymmetric (i.e. different) costs**
  - Like 'cost leadership'
- **Product differentiation/branding**
  - Undercutting the price by a small amount may no longer deliver all of total demand
- **Dynamic competition**
  - What if rivals can retaliate? E.g. you set a high price but threaten to retaliate next time if your rival undercuts you?
- **Capacity constraints**
  - What good is undercutting your rival to get total demand if you cannot actually supply all this demand due to capacity constraints?

## Kreps-Scheinkman (1983): resolving the puzzle

- Two-stage game:
  1. Stage 1: Firms choose capacities (quantities)
  2. Stage 2: Firms compete in prices
- **Result:** Equilibrium outcome = Cournot!
- **Intuition:**
  - Capacity choice commits firms
  - Price competition is constrained by capacity
  - Undercutting is limited by what you can produce
- Key insight: commitment matters

## When does each model apply?

- **Cournot applies when:**
  - Capacity constraints matter
  - Firms commit to production before selling
  - Quantities are hard to adjust quickly
- Examples: manufacturing, airlines (seat capacity)
- **Bertrand applies when:**
  - Prices adjust quickly
  - No capacity constraints
  - Homogeneous products
- Examples: online retail, commodities



## Industry examples: which model?

| Industry          | Model          | Why?                                  |
|-------------------|----------------|---------------------------------------|
| Airlines          | Cournot        | Capacity (planes, gates) committed    |
| Cement            | Cournot        | Production committed; transport costs |
| Gasoline stations | Bertrand       | Prices adjust daily; commodity        |
| Online retail     | Bertrand       | Instant price changes; no capacity    |
| Smartphones       | Differentiated | Product differentiation dominates     |

- In practice: most markets have differentiated products!

## Practice: Cournot vs Bertrand

- **Question:** Two firms have  $MC = 20$ . Market demand is  $P = 100 - Q$ .
- (a) Find equilibrium price under Cournot.
- (b) Find equilibrium price under Bertrand.
- (c) Which model yields higher consumer surplus? Why?

## Practice: Cournot vs Bertrand (solution)

- **Solution:**

- **(a) Cournot:** Using  $P^* = \frac{a+nc}{n+1} = \frac{100+2(20)}{3} = 46.67$

- Or:  $q^* = \frac{100-20}{3} = 26.67$ ,  $Q = 53.33$ ,  $P = 46.67$

- **(b) Bertrand:**  $P = MC = 20$

- **(c)** Bertrand has higher CS:

- Bertrand:  $CS = \frac{1}{2}(100 - 20)(80) = 3200$

- Cournot:  $CS = \frac{1}{2}(100 - 46.67)(53.33) \approx 1422$

- Lower price  $\Rightarrow$  higher quantity  $\Rightarrow$  higher CS

## Oligopoly refresher: Key Points

1. **Cournot:** Firms choose quantities;  $P > MC$
2. **Bertrand (homogeneous):**  $P = MC$ , zero profits
3. **Bertrand paradox:** Only 2 firms but competitive outcome
4. Cournot applies with capacity constraints; Bertrand with flexible prices
5. **Escaping the paradox:** Capacity constraints, differentiation, collusion
6. **Kreps-Scheinkman:** Capacity then price  $\Rightarrow$  Cournot outcome

## Next time

- **Lecture 8:** Product Differentiation and Hotelling
  - Product differentiation: why it matters
  - Hotelling model of spatial competition
  - Connection to demand estimation